

Anirudh Tyagi

📍 Chennai, India ✉ anirudhtyagi188@gmail.com ☎ +91 73032 89966 in itsanirudhtyagi 🌐 anirudh-tyagi

Experience

Software Development Intern

Cadence Design Systems

Noida, India

June 2025 – July 2025

- Worked as a member of Allegro RD team, developing a machine learning-enabled simulator and compiler
- Engineered end-to-end data flow and created Python scripts for data acquisition modules with a web app using Flask, building and optimizing an ML model with 91% accuracy and handled backend infrastructure for secure inference delivery
- Optimized application performance and resolved complex technical issues in a MEAN Stack application for enhancing user experience and functionality

Research Intern

NIOT

Chennai, India

July 2024 – Sept 2024

- Explored 18Tb worth deep-sea metagenome data using bioinformatics techniques including sequence preprocessing including assembly, BLAST analysis, and sequence alignment.
- Worked to identify novel genes and microbial species with potential industrial applications.
- Gained experience in handling and analyzing large, complex and raw, unassembled biological datasets.
- Contributing to the institute's ongoing research efforts in marine biotechnology and Big data-driven genomics.

Education

Sai University

B.Tech in Computing and Data Science

Sept 2022 – May 2026

- Minor: Biological Sciences
- **Relevant Coursework:** Artificial Intelligence, Deep Learning, Reinforcement Learning, ML, DSA, Speech and Image Analysis, Bioinformatics, Software Engineering, Natural Language Processing

Projects

Credit Management Backend(*RESTful API Development, Django, Django REST Framework, Docker, Celery, PostgreSQL, pytest*)

[link-git](#) 📄

- Implemented a credit scoring algorithm that calculates a score from 0-100 based on financial history. The system's loan eligibility engine dynamically corrects interest rates based on credit score bands, with minimums of 12% and 16% for specific score ranges
- Orchestrated a multi-container Docker architecture with a PostgreSQL database and Celery for background tasks, improving scalability and performance
- Developed comprehensive test suites with pytest to ensure application reliability and achieved 100

Cyber Cipher(*RESTful API Development, HTML5 / CSS3, Python, React.js, TypeScript, Tailwind*)

[vercel-app](#) 📄

- Developed CyberCipher, an advanced cryptographic analysis and visualization platform, integrating 2+ multi-algorithm cipher engines (RC4, ChaCha20) and 3+ advanced key generation methods
- Implemented statistic security analysis module, achieving [e.g., 90% accuracy in vulnerability detection] and calculating Shannon entropy
- Designed a modern, responsive UI/UX supporting [e.g., 3+] interactive data visualizations for cryptographic metrics

Kaleidoscope Generator(*Image Processing, Computer Vision, Streamlit, OpenCV*)

[link-git](#) 📄

- Developed a web application that transforms user-uploaded images into symmetrical kaleidoscope patterns with customizable segments
- Implemented core image processing logic using OpenCV and NumPy to perform rotation and mirroring, and applied filters with a selection of psychedelic colormaps
- Built an interactive, user-friendly interface with Streamlit that allows for real-time visualization of the kaleidoscopic effects

Ornamental Flowers Classification(*Transfer Learning, CNNs, Fine-tuning, Image Classification, TensorFlow, Keras*)

[link-CNN](#) 📄

- Preprocessed image data with resizing, normalization, and augmentation techniques improving model performance by 7%
- Implemented and fine-tuned InceptionResNetV2, EfficientNetV2B0, and ConvNeXtTiny models for 96% accuracy
- Optimized classification results through comparative analysis and fine-tuning of CNN models

Skills

Languages: C++, C, Java, Python, MySQL, JavaScript, Django, HTML, R, Julia, Node.js, Flask, TypeScript

Modelling and Analysis: Machine Learning, Natural Language Processing, Deep Learning, Computer Vision, Reinforcement Learning

Software/Platforms: Git, VS Code, Google Colab, AWS, Jira, Notion, Jupyter Notebook, PyCharm, Hugging Face

Technical Skills: DSA, Machine Learning Algorithms, Data Analytics, Data Preprocessing, Deep Learning, NLP Techniques, PyTorch, AI, TensorFlow, Statistical Analysis, API Development, Predictive Modeling, Retrieval-Augmented Generation (RAG), Cloud Computing, Speech and Image Processing

Achievements

Cadence India Hackathon - Top 5 Finalist

2025

- Built BLINC, a decentralized log analysis system integrating, federated learning, machine learning, blockchain (PoS), and IPFS to deliver secure, scalable, and tamper-proof log management
- Achieved prototype-ready deployment with anomaly detection models (XGBoost, CatBoost, LightGBM ensemble) and interactive visualizations for real-time insights

Sports Society President, Sai University

2022

- Spearheaded the organization and execution of intra-school matches, football leagues, and other sports events as the President of the Sports Society, managed budgets, developed food menus, and coordinated with vendors and stakeholders
- Drafted and revised the society's constitution, managed funds and logistics for sports events

Volunteer Swimming Instructor - Taught over 100 individuals

June 2023 - Aug 2024

- Promoted water safety and lifesaving techniques by instructing over 100 individuals of diverse ages, cultivating a safer community around water activities
- Designed and executed personalized lesson plans, successfully improving participants' skills and confidence, leading to a high rate of engagement and positive feedback

Certifications

AI and ML - Cadence

[link](#) 

C++ Language - Cadence

[link](#) 